

コンパイラ演習課題 (2)(解答例)

1. 逆ポーランド記法

(a) $a*b+c$

$ab*c+$

(b) $a*b-c*(d+e)$

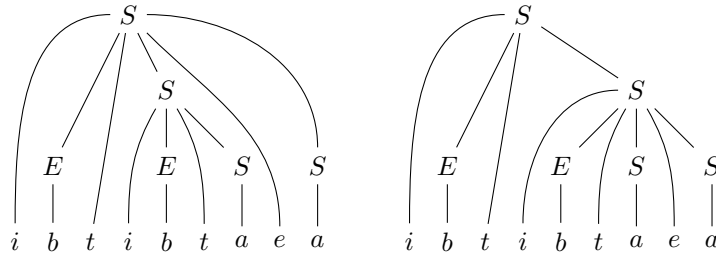
$ab*cde+*-$

2. (a)

$$\left\{ \begin{array}{l} S \rightarrow iEtSS', \\ S \rightarrow a, \\ S' \rightarrow eS, \\ S' \rightarrow \varepsilon, \\ E \rightarrow \end{array} \right\}$$

(b)

$ibtibtaea$ に対して以下の2つの構文解析木が存在する.



(c)

G_1

G'_1

生成規則:

1: $S \rightarrow iEtS$

2: $S \rightarrow iEtSeS$

3: $S \rightarrow a$

4: $E \rightarrow b$

LL(1) 構文解析表:

	i	t	e	a	b	$\#$
S	1 / 2			3		
E					4	

生成規則:

1: $S \rightarrow iEtSS_1$

2: $S \rightarrow a$

3: $S_1 \rightarrow eS$

4: $S_1 \rightarrow \varepsilon$

5: $E \rightarrow b$

LL(1) 構文解析表:

	i	t	e	a	b	$\$$
S	1			2		
S_1			3 / 4			4
E					5	

3.

(a)

$\text{First}(\varepsilon) = \{\varepsilon\}$

$\text{First}(B) = \{w\}$

$$\begin{aligned}
\text{First}(D) &= \{\varepsilon, x, y\} \\
\text{First}(E) &= \{\varepsilon, y\} \\
\text{First}(F) &= \{\varepsilon, x\} \\
\text{First}(S) &= \{u\} \\
\text{First}(v) &= \{v\} \\
\text{First}(w) &= \{w\} \\
\text{First}(x) &= \{x\} \\
\text{First}(y) &= \{y\} \\
\text{First}(z) &= \{z\} \\
\text{First}(Bv) &= \{w\} \\
\text{First}(Dz) &= \{x, y, z\} \\
\text{First}(EF) &= \{\varepsilon, x, y\} \\
\text{First}(BDz) &= \{w\} \\
\text{First}(uBDz) &= \{u\}
\end{aligned}$$

(b)

$$\begin{aligned}
\text{Follow}(S) &= \{\#\} \\
\text{Follow}(B) &= \{v, x, y, z\} \\
\text{Follow}(D) &= \{z\} \\
\text{Follow}(E) &= \{x, z\} \\
\text{Follow}(F) &= \{z\}
\end{aligned}$$

(c) 生成規則に番号をふる.

- 1: $S \rightarrow uBDz$
- 2: $B \rightarrow Bv$
- 3: $B \rightarrow w$
- 4: $D \rightarrow EF$
- 5: $E \rightarrow y$
- 6: $E \rightarrow \varepsilon$
- 7: $F \rightarrow x$
- 8: $F \rightarrow \varepsilon$

下向き構文解析表:

	u	v	w	x	y	z	$\$$
S	1						
B			2 / 3				
D				4	4	4	
E				6	5	6	
F				7		8	

(B, w) に適用ルールの競合があるので, LL1 ではない.

(d) B_{tail} を新たに導入する.

$$P_2 = \left\{ \begin{array}{ll} 1: S \rightarrow uBDz, & 6: E \rightarrow y, \\ 2: B \rightarrow wB_{tail}, & 7: E \rightarrow \varepsilon, \\ 3: B_{tail} \rightarrow vB_{tail}, & 8: F \rightarrow x, \\ 4: B_{tail} \rightarrow \varepsilon, & \\ 5: D \rightarrow EF, & 9: F \rightarrow \varepsilon \end{array} \right\}$$

	<i>u</i>	<i>v</i>	<i>w</i>	<i>x</i>	<i>y</i>	<i>z</i>	\$
<i>S</i>	1						
<i>B</i>			2				
<i>B</i> _{tail}		3		4	4	4	
<i>D</i>				5	5	5	
<i>E</i>				7	6	7	
<i>F</i>				8		9	

(e)

$$\begin{aligned}
 S &\Rightarrow_1 u B D z \\
 &\Rightarrow_2 u w B_{\text{tail}} D z \\
 &\Rightarrow_3 u w v B_{\text{tail}} D z \\
 &\Rightarrow_3 u w v v B_{\text{tail}} D z \\
 &\Rightarrow_4 u w v v D z \\
 &\Rightarrow_5 u w v v E F z \\
 &\Rightarrow_6 u w v v y F z \\
 &\Rightarrow_8 u w v v y x z
 \end{aligned}$$

